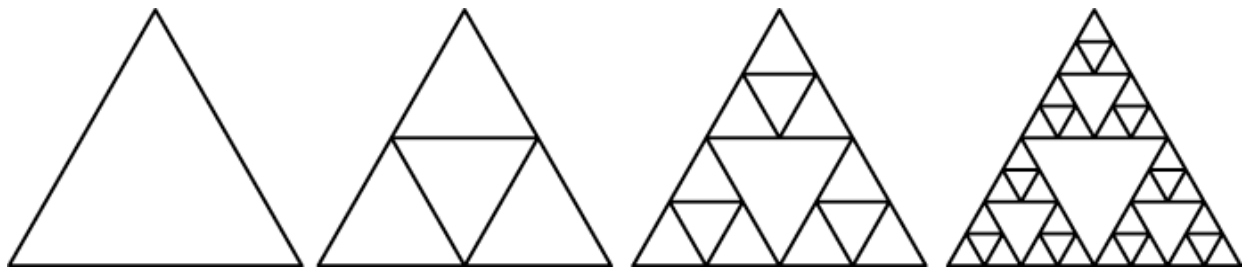


The Sierpiński Triangle

The easiest fractal to draw on paper is the *Sierpiński triangle*, introduced in Chapter 1. This fractal was described by Polish mathematician Waław Sierpiński in 1915 (predating even the term *fractal*). However, the pattern is at least hundreds of years older.

To create a Sierpiński triangle, start by drawing an equilateral triangle—a triangle with equal-length sides, like the one on the left in [Figure 9-3](#). Then draw an upside-down equilateral triangle inside the first triangle, as on the right in [Figure 9-3](#). You’ll form a shape that, if you’re familiar with the *Legend of Zelda* video games, looks like the Triforce.



[Figure 9-3](#): An equilateral triangle (left) with an upside-down triangle added to form a Sierpiński triangle, with additional triangles recursively added

An interesting thing happens when you draw the inner, upside-down triangle. You form three new, right-side-up equilateral triangles. Inside each of these three triangles, you can draw another upside-down triangle, which will create nine triangles. This recursion can continue forever mathematically, though in reality your pen won’t be able to keep drawing tinier triangles.

This property, describing a full object that is similar to a part of itself, is called *self-similarity*. Recursive functions can produce these objects, since they “call” themselves again and again. Practically, this code must hit a base case eventually, but mathematically, these shapes have infinite resolution: you could theoretically zoom in on the shape forever.

Let’s write a recursive program to create the Sierpiński triangle. The recursive `drawTriangle()` function will draw an equilateral triangle, and then recursively call this function three times to draw the inner equilateral